

VR Platforms Comparison Essay

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MI 401 Understanding Virtual Reality

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April 8, 2025

Introduction

Living in a digital and increasingly online modern world, it is important to understand the specific technologies that are altering our reality and helping us navigate the virtual landscapes available to use. In this paper, I will be actively examining how virtual reality (VR) technology—specifically, different platforms created to be used in VR—have the potential to have an impact (both positive and negative) on how we navigate relationships, conflict, and work. To fully comprehend the strengths of VR, I will be discussing the use of VR as a social platform with the use of avatar embodiment, how the interaction of immersion, presence, and empathy affect individuals, and the capabilities of storytelling within VR act as an ability to alter emotions. I am personally interested in learning to what extent this technology can be used as I have a strong passion for writing and storytelling as a whole; understanding if a platform can be used to connect people with themselves or with others—whether it be via socialization, through empathetic experiences, or powerful storytelling—is something I believe to be important. With this paper, I hope to analyze whether or not the means of VR presented in the platforms and the literature reviewed build an argument of meaningful social interaction, empathy, and storytelling ability.

Course Concepts

Firstly, in examining the potential pros or cons in which VR platforms can be used, it will be important to think about the aspect of socialization. As much as VR can often be used or typically seen as a thrill maker, utilizing VR's capabilities of presence—the sense of an individual *feeling* actually in the virtual moment, whether it be socially or physically—can potentially allow

for “people separated by great distances to socialize, collaborate, and conduct business in new ways” (Bailenson, 176). Socialization can especially happen through the embodiment of avatars, something further explored with empathy.

Secondly, considering how users can learn or develop empathy with VR spaces is crucial to understand the capability of a VR platform. As users become engaged in the platform through the technical capabilities of immersion, users can find themselves connecting with the avatars they embody or those around them, feeling strong emotions when existing in the same space as these avatars (Shin, 66). This can have powerful effects, especially as it relates to possible narratives told.

Finally, thinking about how the platforms can be used to storytell and teach is an important idea. Allowing users to embody specific characters or experiences gives way to the ability to transcend the traditional means of framing and composition, considering a wider scope of options in narrative devices. Utilizing VR as a means to provide experiences is strongly related to the idea of stimulating empathy, allowing users to tap into new emotions (Shin, 65).

VR Platforms

I will be examining three VR platforms to analyze a small window of the capabilities VR holds. The first platform I will be examining is Engage. This platform specializes in aiding educational and work environments, creating a social space for students, teachers, and professionals alike to interact. The platform provides numerous tools such as whiteboards, IFX, and avatar customization to provide users with everything they need to be social and productive. The second platform I will be examining is Victory XR. This platform is similar to Engage, with

its focus on educational spaces, but puts a stronger emphasis on demos and previously designed modules for users to experience together. The third platform I will be examining is VRChat. This platform, unlike the previous two, prioritizes social spaces and heavy avatar customization. Users here are capable of selecting from a wide variety of community designed avatars and spaces built.

Research Articles

To aid in my understanding of the course concepts and the platforms detailed above, I have reviewed two pieces of literature that help frame the use of VR. The first piece by Manuela Chessa and Fabio Solari concerns itself with the use of VR as it relates to educational and work spaces. Upon laying out their framing of the Covid-19 pandemic, the experiment detailed examines a small group of participants, dividing them between two groups to engage in a technology course—one group meeting in a VR space, the other in a traditional video conference style. Reviewing the questionnaires' responses, the research team found that those in VR felt a higher sense of presence than those on video calls, but those in the video calls scored higher in comfort of usability than VR. Overall, it appears that the VR technology leads to the potential increase of engagement from participants, but was limited by the specific model being used (Mozilla Hubs). Users felt more present with the environment and their classroom, ultimately engaging in the material more efficiently. Based on participants' responses, the researchers assume the experience would have been better in a higher immersion environment using VR headsets. I find this interesting as it leads me to question how far VR could further be used to truly immerse the user into an environment as comfortable as reality.

The second piece by Donghee Shin is concerned with users entering a state of flow of immersion when interacting with storytelling experiences in VR. The experiment conducted had hosted 200 participants with a variety of experiences using VR. Participants were either experiencing a highly empathetic news narrative or news clips and were either watching on a screen or immersed in a VR headset. The results show that the level of immersion impacts the user's sense of presence directly, while empathy relies more on the user's own abilities. This finding is relevant to understanding the layers to how immersion and empathy interact when experiencing storytelling in VR. I find this dynamic interesting as it considers the ways that technology can not act alone, but requires the will of the user to engage in what they are seeing as to make something out of it—just as all media works.

Comparison

To begin my analysis of VR platforms and whether or not they demonstrate meaningful uses of the concepts I have detailed out, I want to start by considering the way social use and avatars work on these platforms. The best example of this resides in VRChat, where avatar customization is at its best. Users, through a community created list of avatars at their disposal,

VR Platform	Degree of Avatar Customization
VRChat	Strong
Engage	Medium
VictoryXR	Weak

Table comparing the degree of avatar customization on each VR platform. VRChat's continuously growing collection of designs provides a strong degree of customization; Engage has slight customization that is only available outside of the platform's use; VictoryXR has next to none.

are given a wide variety of options to find specific spaces for them to exist in. Users can pull up the main menu and search for worlds and avatars, all uploaded by other users and free to use. As I see it, it is important for

users to find spaces that they connect with and embody avatars they want to, as it will also help

lead them towards becoming more immersed and experience higher amounts of empathy—users with “more absorptive and willing-to-be-immersed personalities are able to become more immersed in VR content by triggering presence and cueing flow into the empathic process” (Shin, 71). Although a lot of immersion and social interaction relies heavily on the individual, VRChat’s range of avatars and social spaces exist to give users more options to slip into a state of spatial presence once they become comfortable, which will lead users to interacting more naturally with other users and experience stronger social presence. The downside of a platform like VRChat is that it does not have the tools, say, as does Engage or VictoryXR, to let users work and find educational spaces; VRChat’s services work mainly as social gatherings without the tools to do much more. Engage, through its whiteboard and drawing tools, and for VictoryXR, its prebuilt educational spaces, provide users help with interacting with each other in a different form of social interaction as opposed to the grounds VRChat facilitates. Both Engage and VictoryXR, though, struggle with creating meaningful social spaces as they lack strong avatar customization; users all appear to look similar and can ultimately strain the meaningfulness of interactions or even the amount of empathy felt, as I will detail next.

The second concept to further analyze is the idea of empathy. As I have mentioned before, VictoryXR’s best feature is its prebuilt educational demos available for use—I believe this to be one of the best cases of VR’s ability to bring out empathy in users. Immersion creates presence, and when a user feels present within the scene, they can connect with the material much stronger than if they were to be just experiencing it as a video or a more distant perspective (Chessa, Manuela, and Fabio Solari, 1246). Similarly, a work environment as demonstrated in Engage’s meeting spaces could help users connect better with their workers, dealing with

different opinions and perspectives. VRChat's social environments can also facilitate the space for users to engage in empathy, through the interaction between unique users. Empathy, overall, can be highly encouraged in VR spaces as individuals can overcome bias by directly observing experiences designed instead of imagining them (Bailenson, 83). Of course, the limitations of all of these platforms lies firstly in the technical ability of immersion (frame rate, graphics, full range of motion) and the individual's capability to experience empathy, but I would argue that all of these platforms are attempting to trend in the direction of providing the tools for users to further work on their emotions, an important experience to aid in the process of storytelling.

The third concept that I want to analyze in regards to how these VR platforms work is storytelling. As I have already largely detailed, while VRChat's social capabilities seems to be the best, VictoryXR has a great catalogue of experiences for users to engage in; between these two, immersive storytelling is very much possible, depending on the context of the narratives you are trying to achieve.

VR Platform	Tools Available for Storytelling
VRChat	Worlds designed by other users
Engage	Whiteboards, drawing functionality, experiences designed by the platform
VictoryXR	Experiences designed by the platform

Table comparing the tools available on each VR platform to aid in storytelling. VRChat has less functionality in terms of interactivity compared to the experiences designed in the two other platforms.

Through educational means of storytelling, VictoryXR will excel; if the goal of the

storytelling is to evoke empathy, VRChat's interactive and active avatar embodiment will work well. It is important to recognize that the ability, as I have stated before, to immerse oneself in a VR experience heavily relies on the user's willingness. "The immersive experience in VR, in which viewers feel they are part of the action, is not bestowed by HMD (or indeed anything related to technological devices), but is reconstructed via user cognition" (Shin, 69). Ultimately, I would argue that these VR platforms all have tools capable of immersing users in meaningful

stories, but it is up to the designers (those creating the narratives) to make creative decisions that benefit the degree of emotion evoked by the art.

Conclusion

After comparing these three VR platforms as they are related to the course concepts and backed by literature reviewed, I have come to a couple conclusions. Ultimately, at its current state, I feel that VR spaces are best used for social spaces, providing users with the tools to interact in unique ways. VRChat demonstrates a great ability for users to embody avatars and interact in a way that would be difficult in reality (i.e. distance limitations and means of presentation), as well as Engage's work tools supporting educational or work environments. While VictoryXR provides great uses of learning in educational demos, it and the other two platforms (Engage and VRChat) don't succeed in centers of high immersion strong enough to support empathy and storytelling in a highly efficient way. Technical limitations hinder immersion, which hinders presence (both spatial and social), which ultimately damages the capabilities of empathy and influence of storytelling in the VR spaces. I am optimistic that, in the future, these platforms will develop more tools of interaction and higher customization for users to feel stronger senses of presence—as presence is the “sine qua non of VR”—ultimately aiding in the capability of meaningful social interactions, empathy felt, and the impact of storytelling (Bailenson, 19). At its current state, the capabilities are limited.

References

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